



Marketing Analytics and
Consumer Psychometry

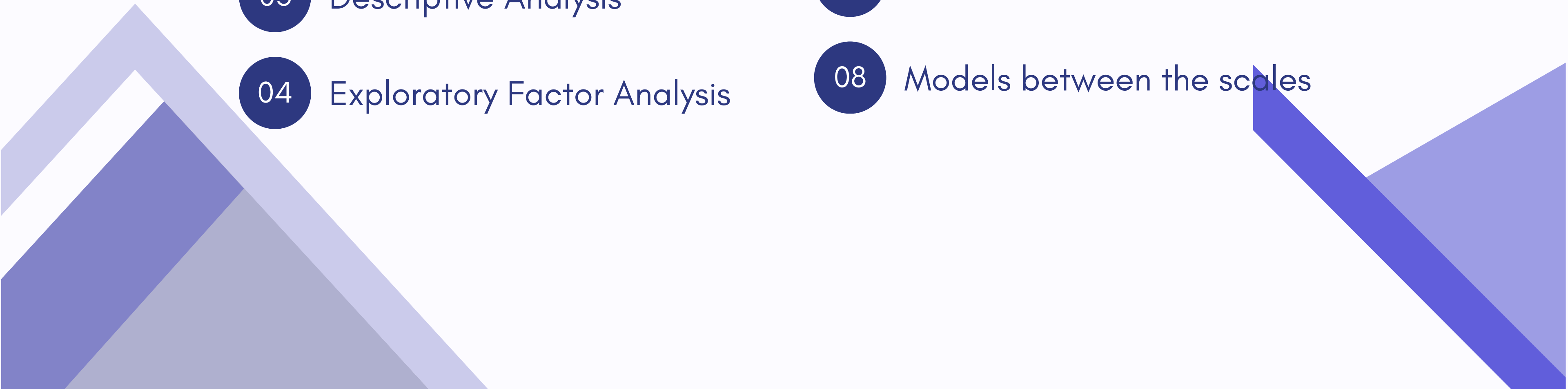
The Fatigue of Digital Context Switching at Work

Scale Development & Validation

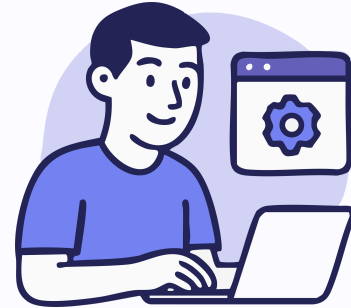
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Agenda Overview

- 01 Introduction
 - 02 The Scale Development
 - 03 Descriptive Analysis
 - 04 Exploratory Factor Analysis
 - 05 Evaluating Items
 - 06 Final items
 - 07 T-TEST AND ANOVA
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Introduction



FFSW, What is it?

- The cognitive and behavioral exhaustion resulting from frequent shifts between different digital work contexts



Why is it interesting?

- Beyond "Screen Time": Unlike general technostress, this targets the structure of activity (switching), not just volume.
- **The Gap:** Intersection of cognitive psychology (attention fragmentation) and occupational health.



The Scale Development

Generating Item Pools

Initial Domains

We identified 3 theoretical dimensions:

- Cognitive Disruption
- Work-Life Intertwining
- Emotional Exhaustion

Initial Pool

Generated 26+ potential items based on literature review.

Dimension 1: Cognitive Disruption in Daily Workflow

- Constant interruptions from notifications make me feel d
- Frequent software switching leaves me mentally confuse
- Switching between multiple apps drains me mentally.
- I lose track of tasks after switching between tools.
- It takes extra time to regain focus after switching tasks.
- I feel mentally overloaded when several tools demand at
- Frequent platform switching slows down decision-making
- I re-read information due to broken concentration.
- Even simple tasks feel harder after rapid switching.
- My thinking feels fragmented after a day of digital switch

Dimension 2: Digital Intertwining of Work and

- I check work messages unconsciously during per
- Work messages during leisure irritate me and ru
- My smartphone blurs work and personal life bo
- I feel guilty not responding immediately to work
- Switching between work and personal apps mak
- It's hard to maintain boundaries between work
- Digital switching disrupts my ability to relax.
- My personal time feels less restorative due to di

Dimension 3: Emotional Exhaustion and Pe

- Managing responses across platforms make
- I feel unproductive due to fragmented work
- Switching between personal and work tasks
- Emotional tone switching across platforms c
- Frequent switching makes me irritable or ur
- I feel tired by midday from constant digital s
- Digital switching makes completing meaning

Please rate how much you agree with the statements about managing tasks at work.

	Strongly disagree	Somewhat disagree	Neutral	Somewhat agree	Strongly agree
It takes me longer to complete the same amount of work.	☐				
I need to recheck work I've completed.	☐				
I miss work deadlines frequently.	☐				
I need to work overtime to complete my regular work.	☐				
I delay responding to non-urgent work messages.					
I use brief and quick responses in communication.					☐
I postpone complex discussions to later times.					☐
I avoid initiating new work conversations.					☐
I proactively set "do not disturb" periods					☐
I batch-process similar work tasks together					☐
I reduce the number of tasks I handle simultaneously.					☐
I get interrupted by notifications while doing focused work.					☐

The Scale Development

Expert Review Process

Behavioral Categories Reviewed

We identified 3 theoretical dimensions:

- 1. Digital Work Behaviors
- 2. Work Efficiency
- 3. Communication Patterns

The Selection Decision:

- Filtered down to 12 Core Items.
- Inclusion Criteria: Items that represent observable behaviors rather than vague feelings.
- Exclusion: Removed items that were too generic

Descriptive Analysis

Descriptives

	N	Missing	Mean	Median	SD	Variance	Minimum	Maximum
DCSF_1	147	0	2.01	2	0.976	0.952	1	5
DCSF_2	147	0	2.78	3	1.197	1.432	1	5
DCSF_3	147	0	1.44	1	0.732	0.535	1	4
DCSF_4	147	0	1.53	1	0.901	0.812	1	5
DCSF_5	147	0	2.52	2	1.289	1.662	1	5
DCSF_6	147	0	3.10	3	1.112	1.237	1	5
DCSF_7	147	0	2.50	2	1.207	1.457	1	5
DCSF_8	147	0	2.19	2	1.112	1.237	1	5
DCSF_9	147	0	2.25	2	1.292	1.669	1	5
DCSF_10	147	0	3.19	4	1.125	1.265	1	5
DCSF_11	147	0	2.78	3	1.127	1.271	1	5
DCSF_12	147	0	2.62	2	1.300	1.689	1	5

Descriptive Statistics Summary:

Sample Size: N = 147

Missing Data: 0%

Key Interpretations:

- Full scale used (1-5) →
- No floor/ceiling issue
- SDs >0 .70 → good variability
- No problematic items

Exploratory Factor Analysis (EFA)

Factor Loadings		
	Factor	
	1	Uniqueness
DCSF_12	0.525	0.724
DCSF_11	0.589	0.653
DCSF_10	0.371	0.862
DCSF_9	0.343	0.883
DCSF_8	0.623	0.612
DCSF_7	0.759	0.424
DCSF_6	0.480	0.770
DCSF_5	0.699	0.511
DCSF_4	0.383	0.853
DCSF_3	0.529	0.720
DCSF_2	0.519	0.731
DCSF_1	0.536	0.713

Exploratory Factor Analysis

Factor Loadings			
	Factor		
	1	2	Uniqueness
DCSF_12	0.411		0.7234
DCSF_11	0.726		0.5494
DCSF_10	0.552		0.7686
DCSF_9	0.388		0.8534
DCSF_8	0.545		0.6299
DCSF_7	0.663		0.4691
DCSF_6	0.590		0.6973
DCSF_5	0.555		0.5250
DCSF_4		0.614	0.6625
DCSF_3		1.090	0.0280
DCSF_2	0.462		0.7231
DCSF_1		0.457	0.6561

Factor Loadings				
	Factor			
	1	2	3	Uniqueness
DCSF_12				0.74632
DCSF_11		0.544		0.54476
DCSF_10		0.790		0.56202
DCSF_9		0.463		0.78727
DCSF_8	0.576			0.61467
DCSF_7	1.268			0.00500
DCSF_6		0.724		0.56298
DCSF_5	0.525			0.50512
DCSF_4			0.632	0.66192
DCSF_3			1.140	0.00500
DCSF_2		0.336		0.73312
DCSF_1			0.408	0.65773

- When conducting factor analyses involving more than one factor, a mathematical error in the model fit was identified.
- Consequently, we employed a single-factor approach to interpret our results.

Evaluating Items

Exploratory Factor Analysis		
Factor Loadings		
	Factor	
	1	Uniqueness
DCSF_1	0.539	0.709
DCSF_2	0.530	0.719
DCSF_3	0.540	0.709
DCSF_4	0.396	0.843
DCSF_5	0.679	0.539
DCSF_6	0.502	0.748
DCSF_7	0.728	0.470
DCSF_8	0.604	0.636
DCSF_9	0.360	0.870
DCSF_10	0.400	0.840
DCSF_11	0.601	0.639
DCSF_12	0.529	0.720

Note. 'Principal axis factoring' extraction method was used in combination with a 'varimax' rotation

[3]

Exploratory Factor Analysis		
Factor Loadings		
	Factor	
	1	Uniqueness
DCSF_1	0.593	0.648
DCSF_2	0.516	0.734
DCSF_3	0.567	0.678
DCSF_5	0.641	0.589
DCSF_7	0.806	0.350
DCSF_8	0.627	0.607

Note. 'Principal axis factoring' extraction method was used in combination with a 'varimax' rotation

[3]

Reliability Analysis	
Scale Reliability Statistics	
	Cronbach's α
scale	0.787

[3]

Item Reliability Statistics	
	Item-rest correlation
DCSF_1	0.513
DCSF_2	0.461
DCSF_3	0.503
DCSF_5	0.551
DCSF_7	0.698
DCSF_8	0.549

Firstly, we found that items such as “batch processing tasks”(Q9~Q12)were actually coping strategies rather than fatigue itself, and were therefore excluded.

Secondly, through factor analysis, we excluded items that performed poorly statistically .

We validated the quality of these six items. Cronbach's Alpha reached 0.787. We constructed a scale with single factor that accurately captures the two core manifestations of '**efficiency loss**' and '**social avoidance**'.

Final items

DCSF_1: It takes me longer to complete the same amount of work.

DCSF_2: I need to recheck work I've completed.

DCSF_3: I miss work deadlines frequently.

DCSF_5: I delay responding to non-urgent work messages.

DCSF_7: I postpone complex discussions to later times.

DCSF_8: I avoid initiating new work conversations.

T-TEST AND ANOVA

Gender:

$p > 0.05$; the null hypothesis cannot be rejected, indicating that the effect between genders is not statistically significant.

Independent Samples T-Test

Independent Samples T-Test						
		Statistic	df	p	Effect Size	
Avg	Student's t	0.304	145	0.762	Cohen's d	0.0501

Note. $H_a \mu_1 \neq \mu_2$

Assumptions

Normality Test (Shapiro-Wilk)		
		p
Avg	0.972	0.004

Note. A low p-value suggests a violation of the assumption of normality

Group Descriptives						
	Group	N	Mean	Median	SD	SE
Avg	1	77	2.26	2.17	0.729	0.0831
	2	70	2.22	2.17	0.810	0.0969

Education and Age Groups:

We separately tested the effects of educational attainment and different age groups on the results using ANOVA. As $p > 0.05$, we likewise cannot reject the null hypothesis.

One-Way ANOVA

Education

One-Way ANOVA					
		F	df1	df2	p
Avg	Welch's	0.648	4	37.4	0.632
	Fisher's	0.666	4	142	0.616

Assumption Checks

Homogeneity of Variances Test (Levene's)				
	F	df1	df2	p
Avg	0.116	4	142	0.977

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One-Way ANOVA

Age

One-Way ANOVA					
		F	df1	df2	p
Avg	Welch's	1.09	4	26.4	0.384
	Fisher's	0.926	4	142	0.451

Assumption Checks

Homogeneity of Variances Test (Levene's)				
	F	df1	df2	p
Avg	1.99	4	142	0.099

[4]

Models between the scales

Mediation: Does social media exacerbate FOMO, thereby leading to impulse purchases?

Hypothesis: Social media usage(SMA) (X) → leads to FOMO (M) → which in turn leads to impulse purchases (OSI)(Y).

Logic: The more people scroll through social media, the more they see of what others possess, thereby generating FOMO anxiety. To alleviate this anxiety, they resort to impulse shopping.

Mediation

Mediation Estimates						
Effect	Label	Estimate	SE	Z	p	% Mediation
Indirect	a × b	0.113	0.0472	2.40	0.017	22.1
Direct	c	0.399	0.0772	5.16	< .001	77.9
Total	c + a × b	0.512	0.0637	8.03	< .001	100.0

The results revealed a significant **indirect effect** of Social Media on Impulsiveness through FOMO (Estimate = 0.113, p = 0.017). However, the **direct effect** of Social Media on Impulsiveness remained statistically significant (Estimate = 0.399, p < .001), indicating a **partial mediation** model.

Models between the scales

Mediation: Is FOMO the cause of information overload, thereby triggering impulse purchases?

Hypothesis: FOMO (X) → leads to shopping information overload (M) → which in turn leads to impulse buying (Y).

Logic: Fear of missing out drives you to frantically browse vast amounts of product information, causing your brain to become overwhelmed. Under pressure and confusion, you become more prone to panic buying.

Mediation

Mediation Estimates

Effect	Label	Estimate	SE	Z	p	% Mediation
Indirect	a × b	0.160	0.0445	3.59	< .001	37.6
Direct	c	0.265	0.0749	3.54	< .001	62.4
Total	c + a × b	0.425	0.0675	6.29	< .001	100.0

- The results showed a **robust partial mediation** effect (Indirect Effect = 0.160, p < .001), accounting for **37.6%** of the total effect. This supports a '**Trait-State-Behavior**' framework.

Models between the scales

Moderation: Is FOMO the cause of information overload, thereby triggering impulse purchases?

Hypothesis: The effect of social media usage (X) on impulse buying (Y) is moderated by FOMO (W).

Logistics: For those prone to FOMO, scrolling through social media can easily trigger impulsive purchases; but for those less susceptible to FOMO (who aren't bothered about missing out), browsing social media might just elicit a chuckle without prompting any buying.

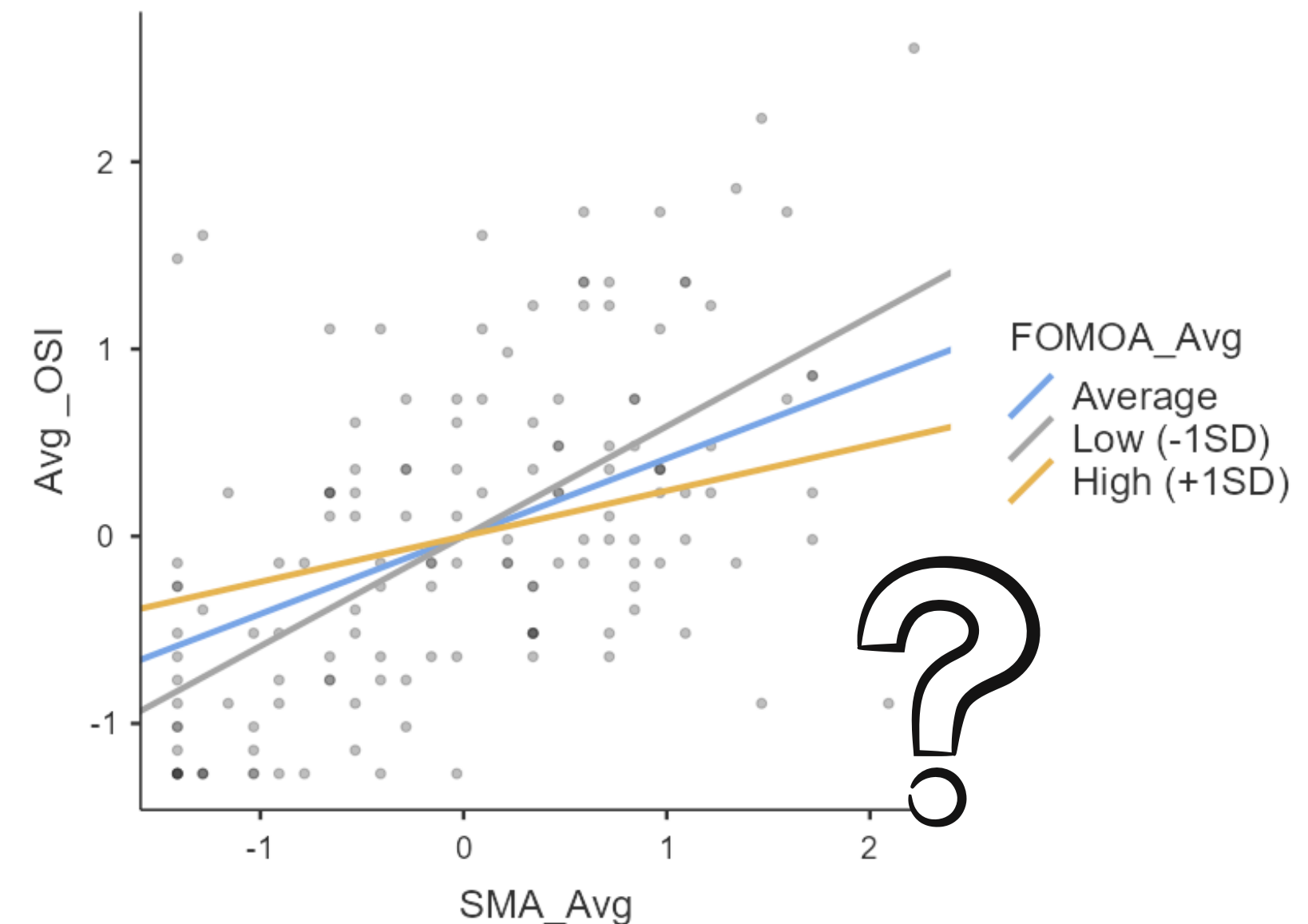
Moderation

Moderation Estimates

	Estimate	SE	Z	p
SMA_Avg	0.415	0.0609	6.82	< .001
FOMOA_Avg	0.154	0.0606	2.54	0.011
SMA_Avg * FOMOA_Avg	-0.184	0.0662	-2.78	0.005

Simple Slope Analysis

Simple Slope Plot





Marketing Analytics and
Consumer Psychometry

Thank You